

KIRAN NASEER

Doctoral Researcher · Federated Learning · Robustness under Distribution Shift · Medical Imaging AI

25016119-003@uog.edu.pk · kirannaseer8@gmail.com · +92 336 6508892 · Gujrat, Pakistan

ORCID · [Google Scholar](#) · [GitHub](#) · [LinkedIn](#)

RESEARCH PROFILE

Doctoral researcher at the University of Gujrat with ten years of university teaching experience. My work asks when AI systems fail the people they were built to serve — the worst-off client in a federation, the hospital whose scanner differs from the rest, the domain a vision–language model was never fine-tuned on. The PhD develops FORGE, a unified framework for diagnosing, benchmarking and stabilising AI under real-world distribution shift, spanning federated learning, parameter-efficient fine-tuning, multimodal robustness and label-efficient medical image segmentation. One paper accepted at FL@FM–IJCAI 2026; three under review — at MDPI Electronics, the Journal of Magnetic Resonance Imaging (JMRI) and IEEE Transactions on Multimedia; one in preparation with an international collaborator at the University of Hyogo (Japan). Invited visiting researcher at HASLab, INESC TEC / Universidade do Minho, Portugal (Oct–Nov 2026). Reviewer for MIUA 2026.

PUBLICATIONS

PEER-REVIEWED — ACCEPTED

When More Parameters Hurt: Foundation Model Priors Amplify Worst-Client Disparity Under Extreme Federated Heterogeneity ACCEPTED

Naseer, K. & Shoaib, U. — FL@FM Workshop, IJCAI–ECAI 2026, Bremen, Germany · [arXiv:2605.08992](#)

Controlled comparison of a task-specific baseline against a PEFT-adapted foundation model across four Non-IID heterogeneity levels, establishing worst-client accuracy as a primary evaluation criterion in federated NLP. Superseded by the journal study below, which reverses the headline direction under a corrected per-client metric and multi-partition design.

UNDER REVIEW

Perplexity Predicts Protection: Choosing Pretrained Backbones for Worst-Client Fairness in Federated Parameter-Efficient Fine-Tuning UNDER REVIEW

MDPI Electronics (Special Issue: Federated Learning and Its Application) · Manuscript electronics-4562418 · submitted 29 August 2026

A backbone-selection rule requiring no federated training: ranking candidates by per-word masked-LM perplexity on a sample of task text reproduces the ranking of their worst-client benefit in all three datasets (Spearman $\rho = -0.867$ across nine cells), validated out-of-sample on a held-out backbone with the interval committed in advance. 313 pre-registered runs across three backbones $\times$ three datasets, five partitions per cell. Shows personalisation (Ditto) recovers only 4–12% of the local-to-federated gap, and that the benefit derives from aggregation rather than frozen features. Code, run manifest, committed predictions and deviation log released publicly.

MedFL-Stress: A Statistically Rigorous Privacy-Aware Robustness Benchmark for Federated Brain Tumour Segmentation Under Cross-Hospital MRI Heterogeneity UNDER REVIEW

Journal of Magnetic Resonance Imaging (JMRI) · Manuscript JMRI-26-1740

Five-seed confidence intervals, differential-privacy analysis and membership-inference resistance across FedAvg, FedProx, FedBN, SCAFFOLD and FedNova on BraTS 2020, with GradCAM interpretability.

Reassessing Global Gradient-Norm Imbalance in BLIP Fine-Tuning Across Physical Domains UNDER REVIEW

with Prof. Dwarikanath Mahapatra, Khalifa University, UAE — IEEE Transactions on Multimedia (Regular Paper) · Manuscript MM-031554

Multi-seed staged fine-tuning protocol that reduces training instability under severe physical domain shift, evaluated across underwater, aerial and radiological imaging domains.

IN PREPARATION

Adaptive ROI-Aware Active Learning for Label-Efficient Glioma MRI Segmentation: Ground-Truth-Free Annotation Windowing IN PREP

with Prof. Essam Rashed, University of Hyogo, Japan

ROI-aware Borda-fusion active learning with an ET-proportional adaptive annotation window and a ground-truth-free proxy cold start (UNet2D, GT correlation 0.945); proof-of-concept validated on UCSF-PDGM, full-dataset scaling in progress.

ACADEMIC SERVICE

Programme Committee Member (Reviewer) — MIUA 2026

30th Medical Image Understanding and Analysis Conference · University College Dublin & Ulster University · July 20–22, 2026

RESEARCH EXPERIENCE

VISITING POSITIONS

Invited Visiting Researcher · HASLab, INESC TEC / Universidade do Minho · Braga, Portugal Oct – Nov 2026

- One-month invited research visit hosted by Dr. Cláudia Vanessa Brito, extending MedFL-Stress to energy-aware federated learning: porting the federated harness to HASLab infrastructure, instrumenting per-round and per-client energy accounting, and constructing the resulting energy–robustness frontier for worst-client performance.

DOCTORAL RESEARCH

Doctoral Researcher · University of Gujrat, Pakistan 2025 – Present

- FORGE-EVAL / federated PEFT: backbone-selection rule for worst-client fairness; 313 pre-registered runs, public code and deviation log. Under review at MDPI Electronics.
- FORGE-BENCH: reproducible federated benchmarks for cross-hospital brain tumour segmentation under MRI heterogeneity; FedAvg, FedProx, FedBN, SCAFFOLD and FedNova on BraTS 2020. Under review at the Journal of Magnetic Resonance Imaging (JMRI).
- FORGE-DIAG / DAMF: vision–language fine-tuning instability under physical domain shift; multi-seed BLEU-4 = 0.292 ± 0.012 on UICD. Under review at IEEE Transactions on Multimedia.

INTERNATIONAL COLLABORATIONS

International Research Collaborator · Prof. Dwarikanath Mahapatra · Khalifa University, UAE 2025 – Present

- Joint journal extension of DAMF on robust multimodal AI and domain adaptation for medical imaging — under review at IEEE Transactions on Multimedia as "Reassessing Global Gradient-Norm Imbalance in BLIP Fine-Tuning Across Physical Domains."

International Research Collaborator · Prof. Essam Rashed · University of Hyogo, Japan 2026 – Present

- ROI-aware active learning for glioma MRI segmentation: adaptive ET-proportional annotation windowing and a ground-truth-free proxy cold start (UNet2D, validation Dice 0.8689, GT correlation 0.945 on UCSF-PDGM).

TEACHING EXPERIENCE

Lecturer, Computer Science · Higher Education Department, Government of Punjab, Pakistan 2016 – Present

- Undergraduate teaching: Operating Systems, Programming Fundamentals, Computer Networks, Network Security.
- Supervised 50+ undergraduate final-year projects annually across AI, software engineering and systems.
- Mentored students in research methodology, academic writing and LaTeX typesetting.

EDUCATION

PhD, Computer Science · University of Gujrat, Pakistan · Coursework CGPA 3.75 / 4.0 2025 – Present

Thesis: FORGE — A Unified Framework for Diagnosing, Benchmarking and Stabilising AI Under Real-World Distribution Shift.

MS, Computer Science · University of Lahore, Pakistan · Thesis: Fault Localization in VANET 2017

Master of Information Technology · University of the Punjab, Pakistan · First Division 2009

BSc, Science · University of the Punjab, Pakistan · First Division 2006

TECHNICAL SKILLS

Languages Python, C++, PHP

Deep Learning PyTorch, HuggingFace Transformers, TensorFlow, scikit-learn

Federated AI FedAvg, FedProx, FedBN, SCAFFOLD, FedNova, FFA-LoRA, Ditto, Flower (flwr), non-IID partitioning, LoRA / PEFT

Multimodal AI BLIP, LLaVA, CLIP, DAMF (custom staged optimisation), VLM robustness analysis

Medical Imaging MONAI, BraTS 2020, UCSF-PDGM, medical NLP, cross-hospital heterogeneity benchmarking, ROI-aware active learning

Statistics Paired testing, Cliff's δ , Holm–Bonferroni correction, bootstrap CIs, pre-registration and deviation logging

Infrastructure CUDA, Docker, Git, LaTeX, Linux, Kaggle GPU / HPC

REFERENCES

Dr. Nauman Riaz Chaudhry — PhD Supervisor

Department of Computer Science, University of Gujrat, Pakistan · nauman.riaz@uog.edu.pk · +92 300 950 3200

Dr. Naveed Anwer Butt — PhD Co-Supervisor

Department of Computer Science, University of Gujrat, Pakistan · naveed@uog.edu.pk · +92 300 645 2360